



State space model :

The general form of a state space model is as follows :

$$\left. \begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx + Du \end{aligned} \right\} \begin{array}{l} \text{Continuous} \\ \text{time} \end{array}$$

$$\left. \begin{aligned} x[k+1] &= Ax[k] + Bu[k] \\ y[k] &= Cx[k] + Du[k] \end{aligned} \right\} \begin{array}{l} \text{discrete} \\ \text{time} \end{array}$$

x is a vector and its elements are called the "states" of the system.

A is a matrix with dimension $n \times n$. It is referred to as the "system matrix" or "state transition matrix"

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} \quad A = \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \vdots & \vdots & & \vdots \\ A_{n1} & A_{n2} & & A_{nn} \end{bmatrix}$$

$$\dot{x} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \vdots \\ \dot{x}_n \end{bmatrix}$$

B = input matrix
(dimension $n \times 1$)
(column vector)

$C =$ output matrix. Its dimension depends on which states we want to choose. (see example)

D is the direct transmission matrix. It is usually zero.

Let's see an example with a second order differential equation that we have seen before, our beloved spring-mass-damper system.

$$m\ddot{x} + c\dot{x} + kx = f(t)$$

$$\Rightarrow \begin{matrix} \downarrow & \downarrow & \downarrow \\ \dot{x}_2 = \ddot{x} & \dot{x} = \dot{x}_1 = \dot{x}_2 & x = x_1 \end{matrix}$$

$$\begin{cases} \ddot{x} = -\frac{k}{m}x_1 - \frac{c}{m}x_2 \\ \dot{x}_2 = -\frac{k}{m}x_1 - \frac{c}{m}x_2 \end{cases}$$

$$\dot{x}_1 = 0 \cdot x_1 + 1 \cdot x_2 + 0 \cdot f(t)$$

$$\dot{x}_2 = -\frac{k}{m}x_1 - \frac{c}{m}x_2 + \frac{f(t)}{m}$$

$$\Rightarrow \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} f(t) \quad \text{"} u(t) \text{"}$$

$$\Rightarrow \dot{X} = AX + Bf(t)$$

where

$$\dot{X} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} \quad A = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix}$$

$$y = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} u(t)$$

$$C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 0 \end{bmatrix}$$

We will see several other examples as we move forward.

